Computer Communication and Networks

OSI Reference Model

Session 2

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The OSI Reference Model

Learning Outcome:- At the end of the lesson, you are expected to be able to:

1. Sketch on OSI reference model.
2. Identify 7 layers of OSI model.
3. Underline the need for Layering.
4. Explain the encapsulation
5. Describe the data communication.



The OSI Model: Introduction

Introduction:

- Emergence of different hardware and software vendors.
  - ✓ Standardization was needed for interoperability
- In1947, **International Standard Organization** (ISO)-a multinational body dedicated to world wide agreement on international standard was established.
- In 1977, ISO established a subcommittee to develop communication architecture called **Open Systems Interconnection** (OSI) reference model.
- **OSI model is not a protocol**.



The OSI Model: purpose

- **Purpose of OSI model**:
  - To show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software.

    - ✓ allows interoperability across all types of computer systems.
      - allows any two different systems to communicate regardless of their underlying architecture.
        - Dell, IBM, Microsoft systems.

    - ✓ for understanding and designing a network architecture that is flexible, robust, and interoperable.

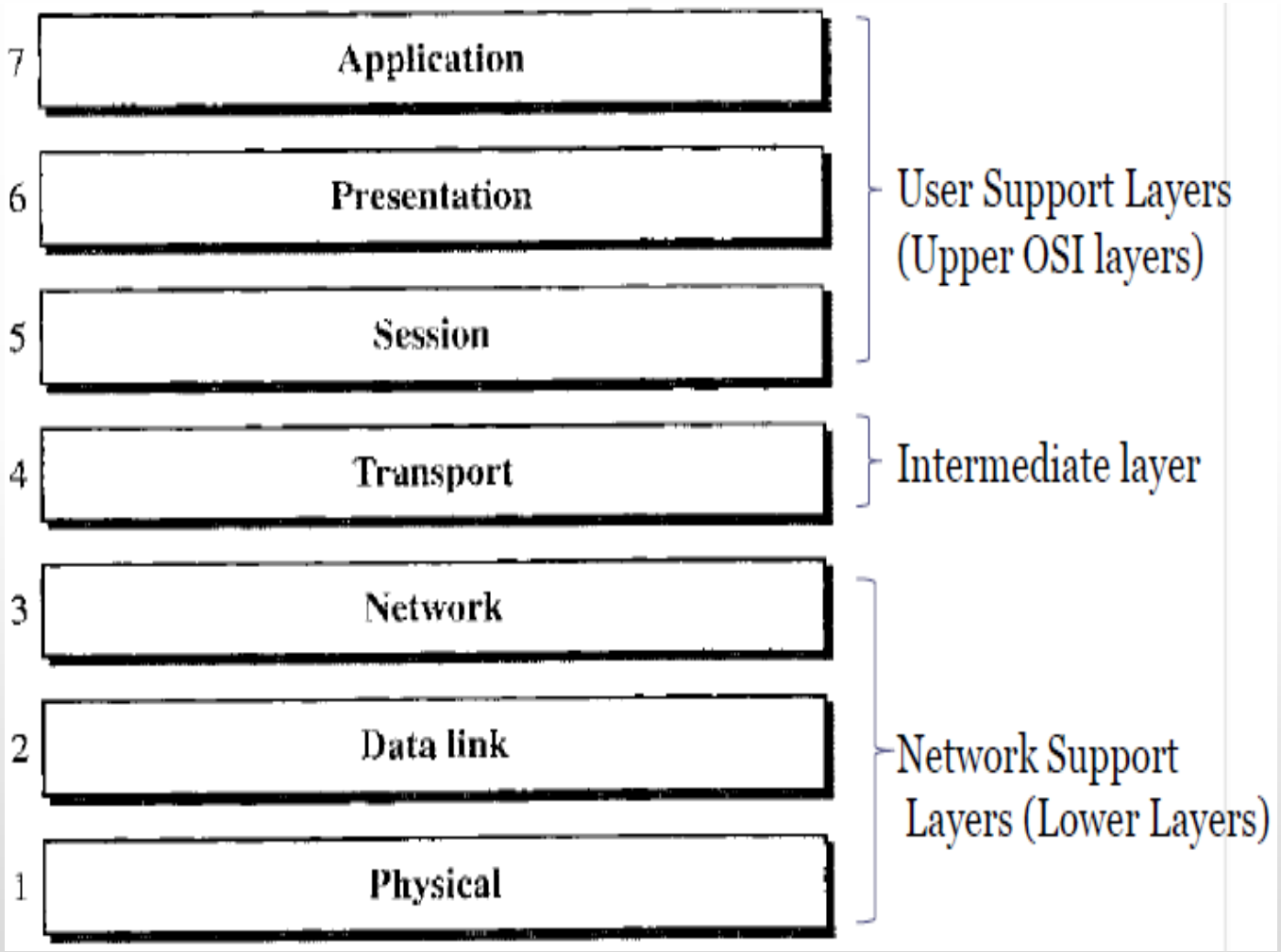
    - ✓ is layered architecture and its ordered.



The OSI Model: 7 Layers

7 layers of OSI model:

- It consists of 7 separate but related layers.
- Each layer relies on the next lower layer and provide service to the next higher layer.





The OSI Model: organization of layers

Organization of layers:

- User support layers (Layer 5, 6, 7):- Upper Layers
  - ✓ Allow interoperability among unrelated software systems.
  - ✓ Almost use software only

- Network support layers (Layer 1, 2, 3):- Lower Layers
  - ✓ Deals with physical aspects of moving data from one device to another.
  - ✓ Used combination of hardware and software

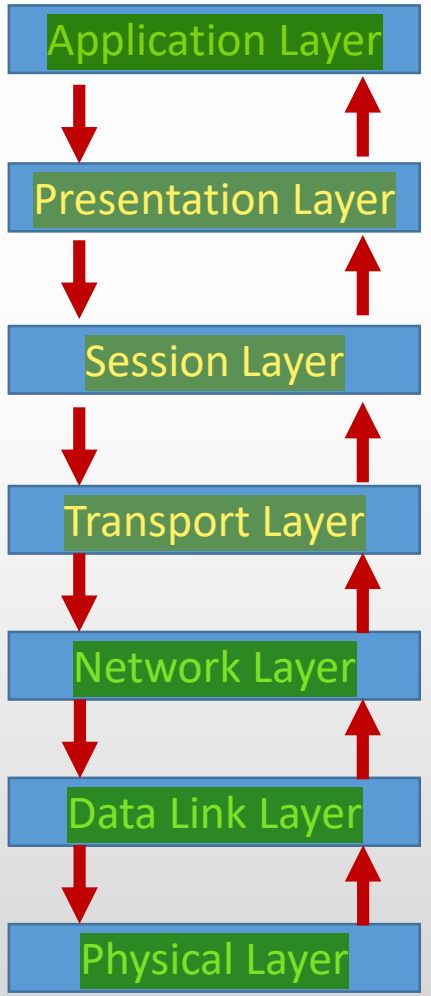
- Intermediate layer (Layer 4):
  - ✓ Links upper layers and lower layers subgroups.
  - ✓ It ensures that delivering of exact data format from lower to upper layers is successful



The OSI Model: Layering

❖ Layering:

- ✓ To decompose communication into set of smaller, well-defined components.
- ✓ Components build on top of one another (layered).
- ✓ Each layer has a well-defined interface & clear responsibilities.
- ✓ Each layer can work-out independently





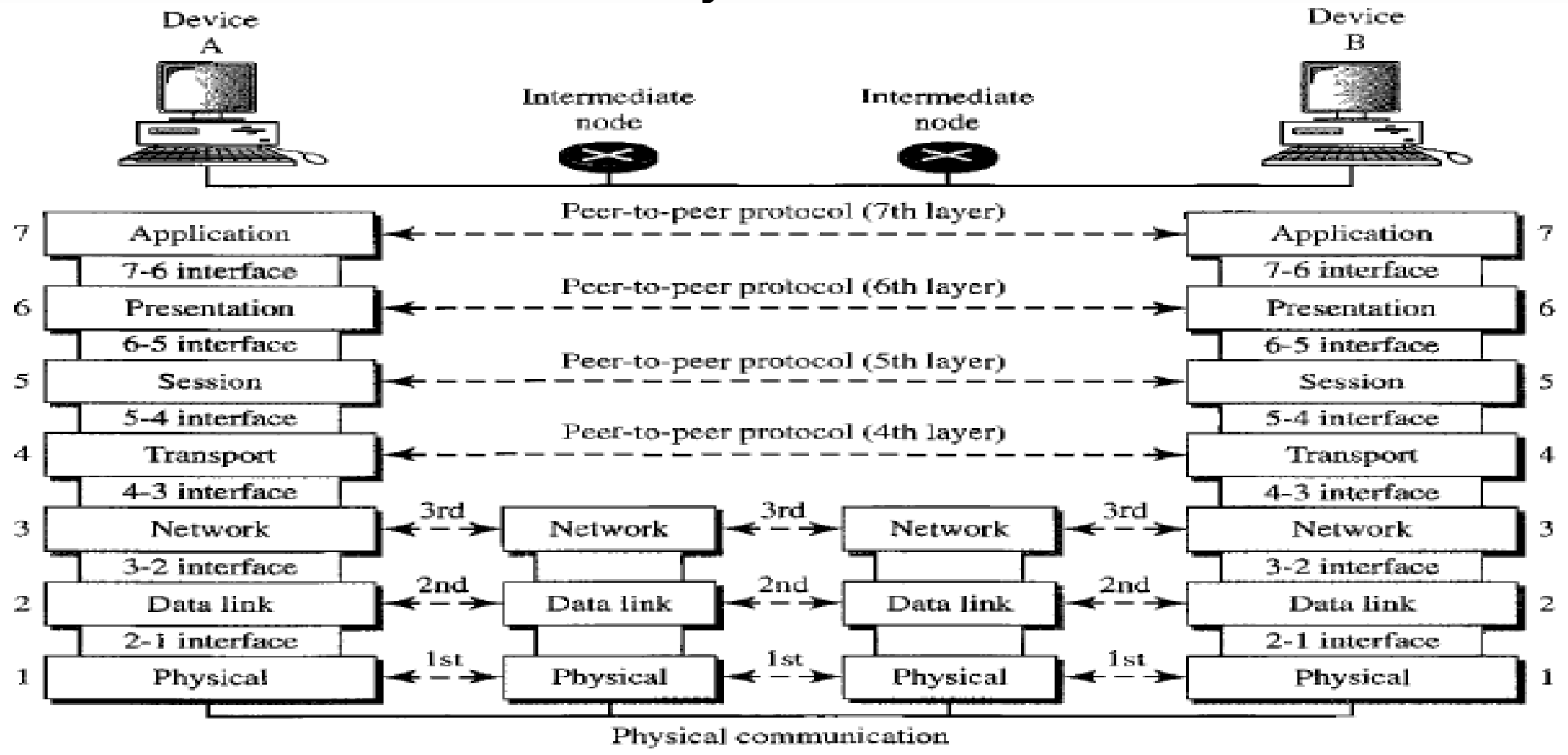
The OSI Model: Principle of Layering

❖ Principle of Layering:

- ✓ Data format.
- ✓ Packetization.
- ✓ Reliability, error checking.
- ✓ Congestion and flow control.
- ✓ Packet delivery and routing.
- ✓ Link delivery
- ✓ Signal Modulation and framing.

The OSI Model: The interaction between layers

❖ The interaction between layers:





The OSI Model: Interaction between layers

Peer-to-peer Process:

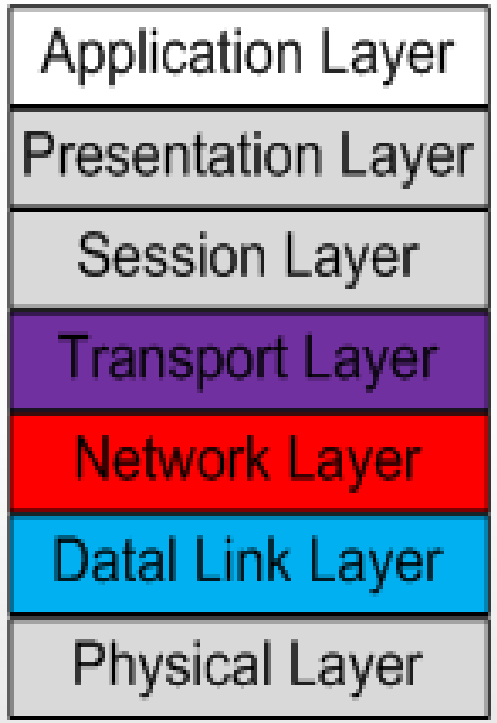
- Lets take an example of communicating machine A and machine B.
  - ✓ Within single/individual machine, each layer depends upon the service of layer just below it. For example, Layer 3 uses service provided by Layer 2 and it provides service to Layer 4.
  - ✓ Between machines, layer n on one machine communicates with layer n on another machine.
    - Communication is governed by an agreed-upon series of rules and conventions called protocol.
- The process of each machine that communicate at a given layer is called Peer-to-Peer Process.
- Each interface defines the information and services a layer must provide for the layer above.



The OSI Model: Encapsulation

- **Encapsulation:**

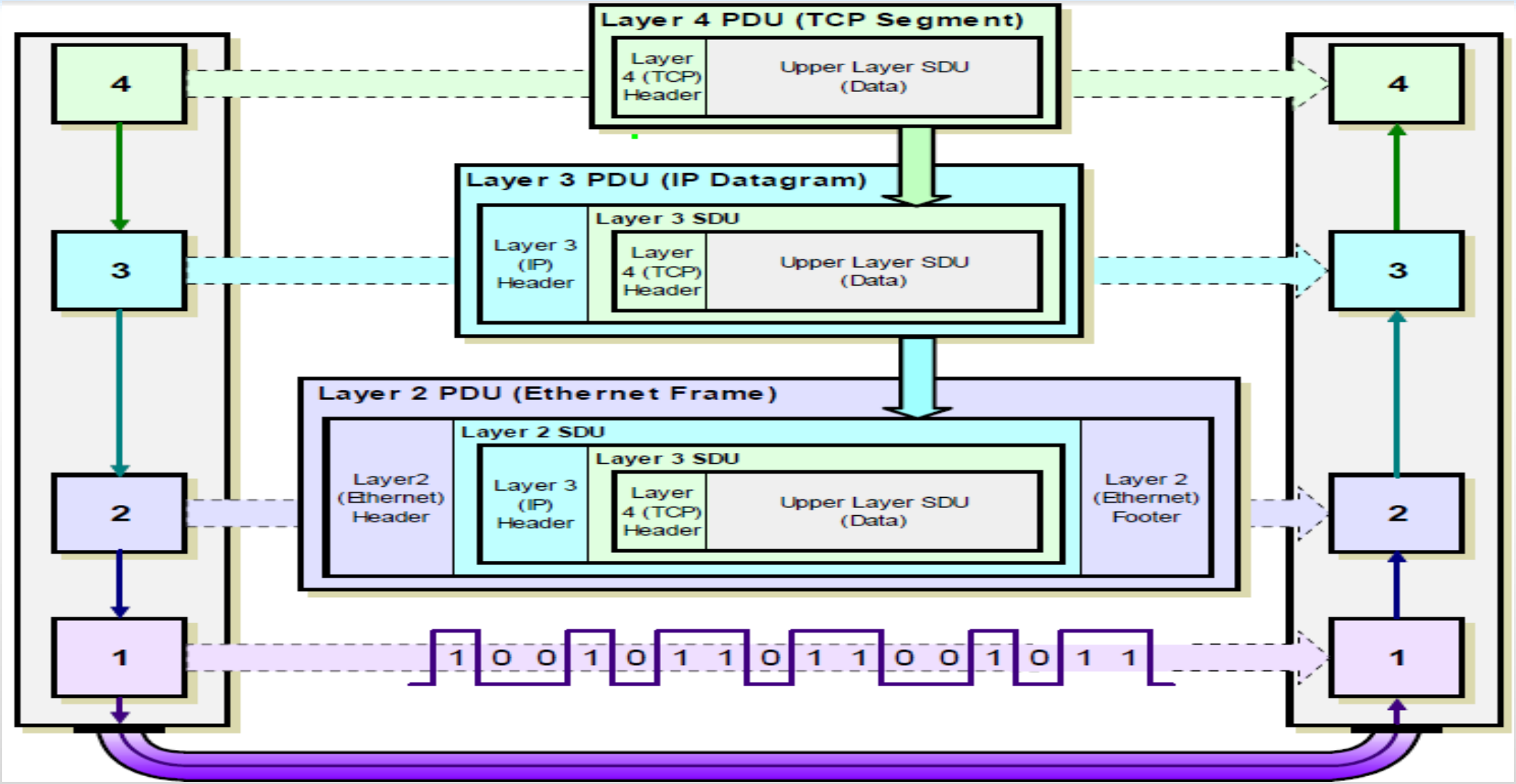
- A packet (header & data) at level 7 is encapsulated in a packet at level 6. The whole packet at level 6 is encapsulated in a packet at level 5 and so on. This concept is called Encapsulation.



- Example:
  - ✓ HTTP (Web) application payload in
  - ✓ A TCP transport segment in
  - ✓ An IP network packet in
  - ✓ An Ethernet link frame



The OSI Model: Encapsulation

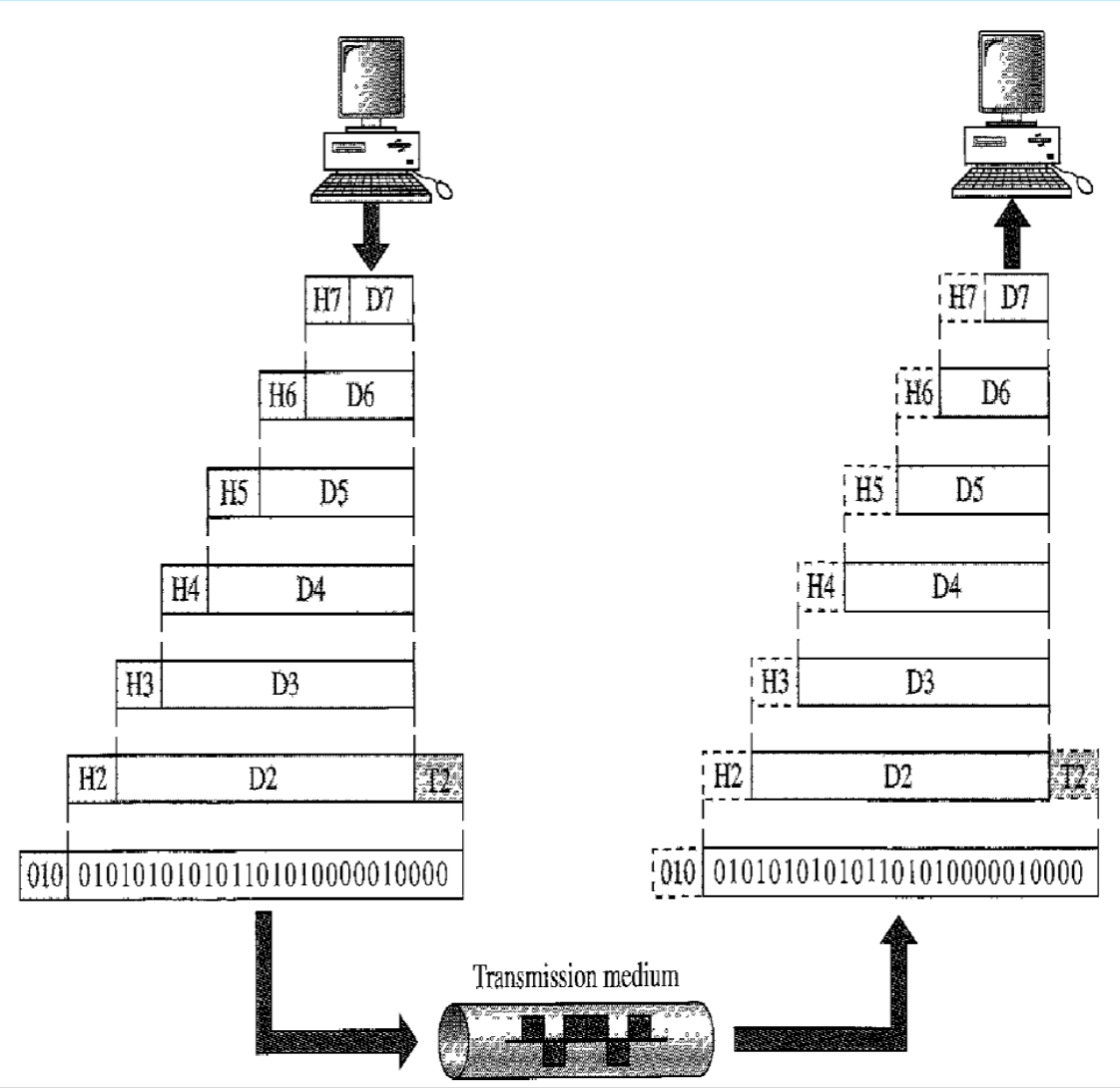




The OSI Model: Data Exchange

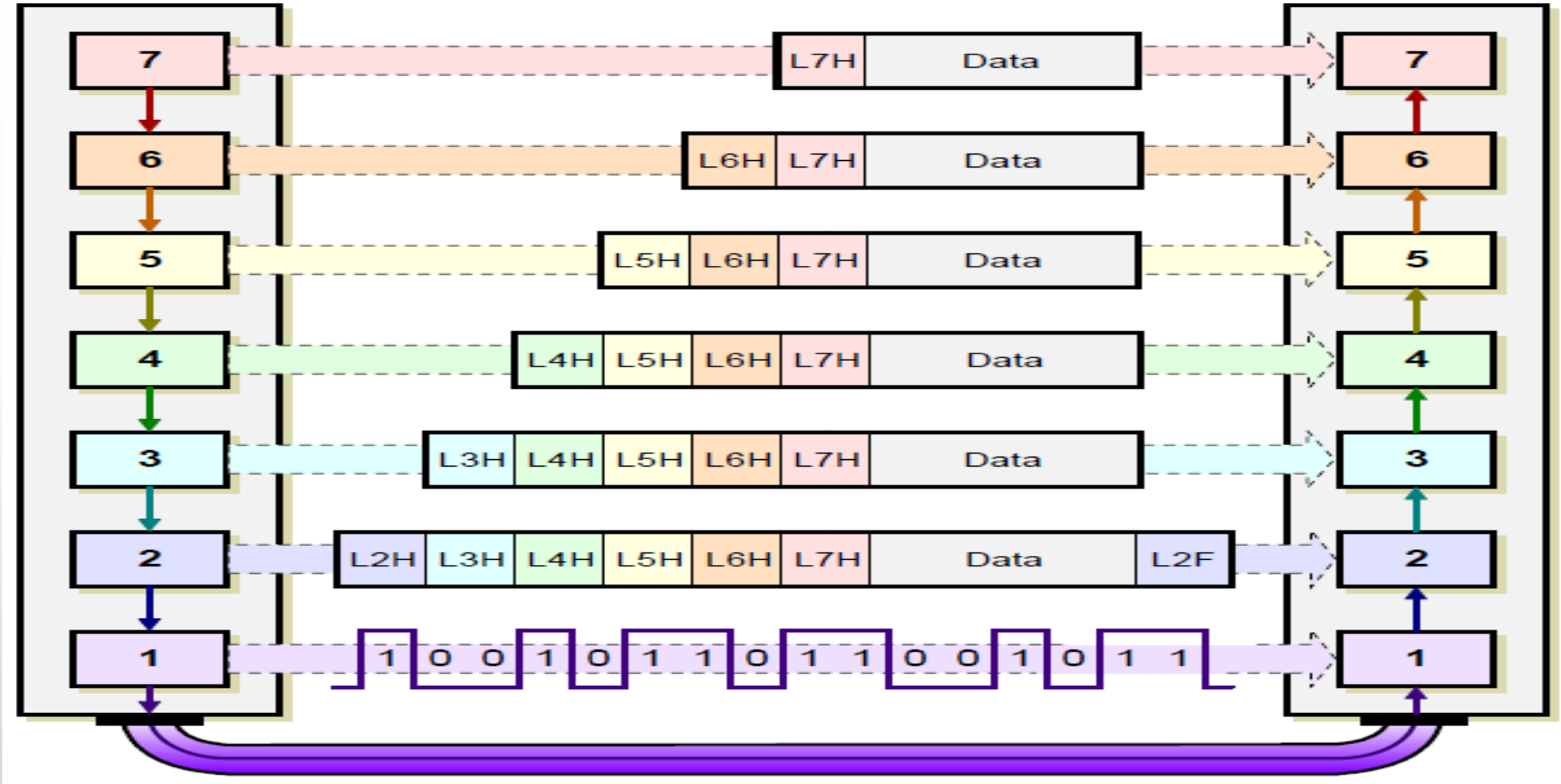
❖ Data Exchange:

- D7 means Data unit at layer7 . D6 means Data unit at layer 6 and so on.
- Header and Trailer is added. **Trailer is added only in Layer 2.**
- When formatted data unit is reached at Physical layer, it changed into electromagnetic signal and transported along a physical link.
- Upon reaching destination, signal is transformed back to bits.





The OSI Model: Data Exchange





The OSI Model: Data Communication

❖ Data Communication:

- When we communicate, we are sharing information. This sharing can be local or remote.
- Between individuals, local communication usually occurs face to face, while remote communication takes place over distance.
- The term telecommunication, which includes telephony, telegraphy, and television, means communication at a distance



The OSI Model: Data Communication

Data Communication:

- Data communications are the exchange of data between two or more devices in the computer network.
- The effectiveness of a data communications system depends on four fundamental characteristics:
  1. Delivery
  2. Accuracy,
  3. Timeliness
  4. Jitter



The OSI Model: Data Communication

- **Data Communication:**
  - **Delivery:** The system must deliver data to the correct destination. Data must be received by the intended device or user.
  - **Accuracy:** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.



The OSI Model: Data Communication

- **Data Communication:**
  - **Timeliness:** The system must deliver data in a timely manner. Data delivered late are useless.
  - **Jitter:** Jitter means the variation in the packet arrival time. It is the uneven delay in the delivery.
  - The five components of data communication are: Message, Sender, Receiver, Transmission Media and Protocol.



Recapitulation

We have learned:

1. Emergence of OSI Model
2. 7 layers of OSI Model
3. Purpose and Layering architecture of OSI Model
4. Data communication from source to destination stations.

Important
1.OSI layer- 7 layers
2,understand Layers
3. Data



References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan

